

Derivative of $f(x) = (x+3)^3$ Using the Limit Definition

We use the definition:

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

1. Compute $f(x+h)$:

$$\begin{aligned} f(x+h) &= (x+h+3)^3 = (x+3+h)^3 \\ &= (x+3)^3 + 3(x+3)^2 h + 3(x+3) h^2 + h^3 \end{aligned}$$

2. Form the difference quotient:

$$\begin{aligned} &[f(x+h) - f(x)] / h \\ &= [(x+3)^3 + 3(x+3)^2 h + 3(x+3) h^2 + h^3 - (x+3)^3] / h \end{aligned}$$

3. Simplify:

$$\begin{aligned} &= [3(x+3)^2 h + 3(x+3) h^2 + h^3] / h \\ &= 3(x+3)^2 + 3(x+3)h + h^2 \end{aligned}$$

4. Take the limit as $h \rightarrow 0$:

$$f'(x) = 3(x+3)^2$$

Final Answer:

$$f'(x) = 3(x+3)^2 = 3x^2 + 18x + 27$$